ARTICLE | Article in Press

Isometric knee strength and postural control show no robust baseline associations in young soccer players

Rafał Studnicki

Received: 26 June 2026 / Accepted: 27 August 2026
© The Author(s) 2026

Abstract

To examine baseline relationships between isometric knee-flexion and knee-extension pad force, the protocol-specific flexion-to-extension force ratio, and force-platform-derived postural-control outcomes in academy soccer players aged 9–12 years, and secondarily to characterise exploratory within-player differences between two assessments conducted across a 5-day soccer camp. This single-centre observational study comprised a cross-sectional baseline component and a single-group repeated-session component. Data were prospectively collected, although study registration was retrospective. Seventy players were included (62 male and 8 female; age, 10.53 ± 1.24 years; maturity offset, −3.78 ± 1.11 years). Knee-flexion and knee-extension pad force were assessed using a ForceFrame isometric dynamometer, whereas jump, landing, stabilisation, centre-of-pressure, and interlimb-asymmetry outcomes were obtained using ForceDecks force platforms. Baseline partial Spearman correlations were adjusted for age group, sex, baseline body mass, and age-group-centred maturity offset. Repeated-session differences were analysed using outcome-specific complete pairs, paired tests, mean differences, Cohen's d_z, bootstrap 95% confidence intervals, and sensitivity analyses. Benjamini–Hochberg false-discovery-rate correction was applied across 18 baseline associations and globally across 24 repeated-session outcomes. Exploratory analyses examined whether maturity offset was associated with T1–T0 differences. Baseline analysis samples comprised 62–64 players, and repeated-session complete-pair samples ranged from 60 to 65. Adjusted baseline coefficients ranged from −0.194 to 0.327, and none remained statistically robust after multiplicity correction (all $q \geq 0.162$). The largest estimate was observed between knee-extension pad force and single-leg-standing centre-of-pressure mean velocity ($\rho = 0.327$, 95% CI 0.059 to 0.557; $p = 0.012$; $q = 0.162$). Knee-extension maximum pad force was 28.86 N higher at T1 (95% CI 16.62 to 41.09), corresponding to 16.3% of the paired-sample T0 mean and d_z = 0.60 (95% CI 0.33 to 0.81; $p < 0.001$; $q < 0.001$). This was the only repeated-session outcome that remained robust after correction across all 24 outcomes. Knee-flexion pad force, the flexion-to-extension force ratio, stabilisation time, and centre-of-pressure mean velocity showed unadjusted differences, but none remained robust after multiplicity correction. In exploratory analyses, age-group-centred maturity offset was associated with T1–T0 differences in bilateral land-and-hold time to stabilisation ($\rho = 0.484$, 95% CI 0.268 to 0.660; $q = 0.002$) and single-leg land-and-hold time to stabilisation ($\rho = 0.380$, 95% CI 0.124 to 0.585; $q = 0.043$). No baseline association between the selected knee-force and postural-control measures was statistically robust after adjustment for developmental and anthropometric covariates and correction for multiple testing. Knee-extension pad force was higher at the

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.

https://doi.org/10.1038/s41598-026-69517-z

Published online: 19 September 2026

scientific reports

second assessment, but the uncontrolled design prevents attribution of this difference to the soccer camp and does not exclude familiarisation, measurement variability, recovery status, or other time-dependent influences. Associations with anthropometrically estimated maturity offset indicated heterogeneity in repeated-session stabilisation-time differences but should not be interpreted as causal effects of maturation.

Trial registration: Retrospective study registration ClinicalTrials.gov ID NCT06719752 registered 30 November 2024.

Keywords Soccer · Child · Muscle strength · Postural balance · Hamstring-to-quadriceps ratio

Introduction

Soccer participation during childhood involves repeated sprinting, jumping, landing, kicking, braking, and changes of direction, placing substantial demands on the developing lower-limb neuromuscular system[1,2]. Growth and biological maturation are important when interpreting neuromuscular test scores in youth soccer because players of the same chronological age may differ in body dimensions, strength, coordination, and postural-control performance[1–3].

Knee flexor and extensor strength contribute to force production and dynamic knee control during soccer-specific actions. In youth soccer, hamstring and quadriceps strength generally increase with age, whereas the hamstring-to-quadriceps ratio varies with developmental stage and testing method[4,5]. This ratio describes relative posterior-to-anterior thigh capacity, but its interpretation depends on contraction mode, joint position, angular velocity, and measurement protocol[6]. Postural control and landing performance also depend on sensory integration, intersegmental coordination, and task-specific force regulation. Static sway, dynamic stabilisation, landing kinetics, and maximal isometric strength are therefore non-equivalent constructs, and associations between balance and lower-limb strength are generally small[7,8].

Because force production, static sway, dynamic stabilisation, landing force, and interlimb asymmetry represent different measurement constructs, their empirical relationships should be established before the measures are combined or interpreted as interchangeable[7]. Validation studies support ForceFrame for selected isometric knee-strength configurations and ForceDecks for force, balance, and movement outcomes in adult samples, but measurement properties remain outcome-, protocol-, and population-specific[9,10]. As emphasized by a previous study[11], sensor-derived assessments require explicit standardisation to permit reproducible interpretation across athletes and settings. Interpretation of portable dynamometry and force-platform outcomes depends on standardised positioning, familiarisation, trial selection, and signal-processing procedures, and measurement properties cannot be assumed to transfer across tasks or populations[9,10]. These controls are particularly important in youth, in whom developmental heterogeneity may increase protocol-dependent variability[12]. The primary knowledge gap concerns whether protocol-specific isometric knee pad-force measures are associated with distinct static-sway, dynamic-stabilisation, landing-force, and asymmetry outcomes in preadolescent soccer players. Previous work in older youth players related isokinetic knee strength to a global stability index, but it did not address the present ForceFrame configuration or the individual ForceDecks constructs examined here[10,13]. A separate, secondary gap concerns the extent to which these scores differ between two closely spaced assessments; such differences may reflect familiarisation, fatigue or recovery, and measurement variability rather than training adaptation.

Most evidence for neuromuscular adaptation in youth concerns structured, progressive training conducted over several weeks rather than 5 days of predominantly technical soccer practice[14,15]. Pre-to-post differences across a 5-day camp should therefore not be interpreted as chronic adaptation. They may reflect transient fatigue or recovery, task familiarisation, day-to-day biological variation, measurement error, or an early exposure-related response. Acute soccer exposure can alter postural-control performance in young players, making the interval between the final training session and post-camp testing methodologically important[16–18]. Accordingly, the primary objective was to assess baseline relationships between knee flexor and extensor strength, the hamstring-to-quadriceps ratio,

and ForceDecks-derived balance, landing-kinetic, and load-asymmetry outcomes in soccer players aged 9–12 years. The secondary, exploratory objective was to characterise short-term within-player pre-to-post response patterns across the 5-day camp, without attributing them to chronic training adaptation or injury-risk modification.

Methods

Study design and analytical framework

This was a single-centre observational field study in which data were collected prospectively from academy soccer participants during 20–24 February 2023. The study comprised two analytically distinct components. The primary component was a cross-sectional baseline analysis of relationships between ForceFrame-derived isometric knee-flexion and knee-extension pad-force measures and selected ForceDecks-derived static-sway, dynamic-stabilisation, landing-force, and interlimb-asymmetry outcomes. The secondary component was an exploratory single-group repeated-session analysis of within-participant differences between the baseline assessment (T0) and a second assessment (T1) conducted after the observed 5-day camp period.

The two components were evaluated in the same cohort because the same standardised assessment battery provided, at T0, information on the convergence of distinct neuromuscular constructs and, at T1, information on short-interval score variation under repeated field-testing conditions. The components were analysed and interpreted separately: the baseline association analysis was the primary scientific question, whereas the repeated-session analysis was secondary and exploratory. The academy camp was an existing activity observed by the investigators and was not assigned or manipulated as a research intervention. There was no randomisation, allocation procedure, non-camp comparator, or counterfactual condition; consequently, T1-T0 differences were not interpreted as causal effects of the camp. Reporting followed the Strengthening the Reporting of Observational Studies in Epidemiology statement (von Elm et al. 2007).

Ethics, consent, assent, and retrospective study registration

The study was approved by the Independent Bioethics Committee for Scientific Research at the Medical University of Gdansk, in cooperation with the AZS Central Academic Sports Center in Gdansk (approval date, 6 February 2023; Resolution No. NKBBN 1/2023) and was conducted in accordance with the Declaration of Helsinki and applicable national requirements for research involving minors.

Before enrolment, a parent or legal guardian received written study information and provided written informed consent. Each participant received an age-appropriate explanation of the procedures and provided written assent. Participation was voluntary, and consent or assent could be withdrawn without consequences for academy participation.

The study was registered retrospectively on 30 November 2024 under ClinicalTrials.gov identifier NCT06719752, approximately 21 months after completion of data collection. This record is reported as retrospective study registration rather than prospective trial registration. Because registration occurred after enrolment and data collection, it does not demonstrate prospective specification of the objectives, outcomes, or analyses. The analyses were retrospectively specified and exploratory.

Setting, recruitment, eligibility, and participant flow

Testing was conducted in the laboratory of the AZS Central Academic Sports Center in Gdansk, Poland, during an academy soccer-camp stay. Participants were recruited from the Varsovia Football Academy and belonged to U10-U12 squads, participating in official league competitions organised by the Masovian Football Association (MZPN) at the provincial level.

Eighty-four academy participants were assessed for eligibility. Fourteen were excluded before enrolment, with the following mutually exclusive reasons: 4 no patient card; 8 injuries; 2 refusals. Seventy participants were enrolled and completed at least part of the T0 assessment. Outcome-specific valid samples differed because of missing or invalid measurements, as described below and in Fig. 1.

Participants were eligible when they were in the recorded 9–12-year age groups, actively participating in academy soccer, medically able to undertake maximal testing, and had parental or guardian consent and participant assent. Exclusion criteria were previous surgery involving the lumbar spine or lower limb; hip, knee, or ankle injury during the preceding 6 months; current lower-limb pain; a neurological condition; a connective-tissue disorder; lower-limb joint hypermobility; and if they had participated in regular, structured resistance training within the 12 months prior to the study. Resistance training was defined as planned exercise with progressively applied external resistance, performed to improve strength capacity. Standard soccer training and natural general development exercises implemented as part of sports training were not classified as resistance training.

Joint hypermobility was assessed using screening test based on training documentation and parents' medical information. This raises the possibility of misclassification as it was not a diagnosis of hypermobility. Eligibility and the ability to complete maximal testing were determined by the principal investigator, a physiotherapist with 25 years of professional experience.

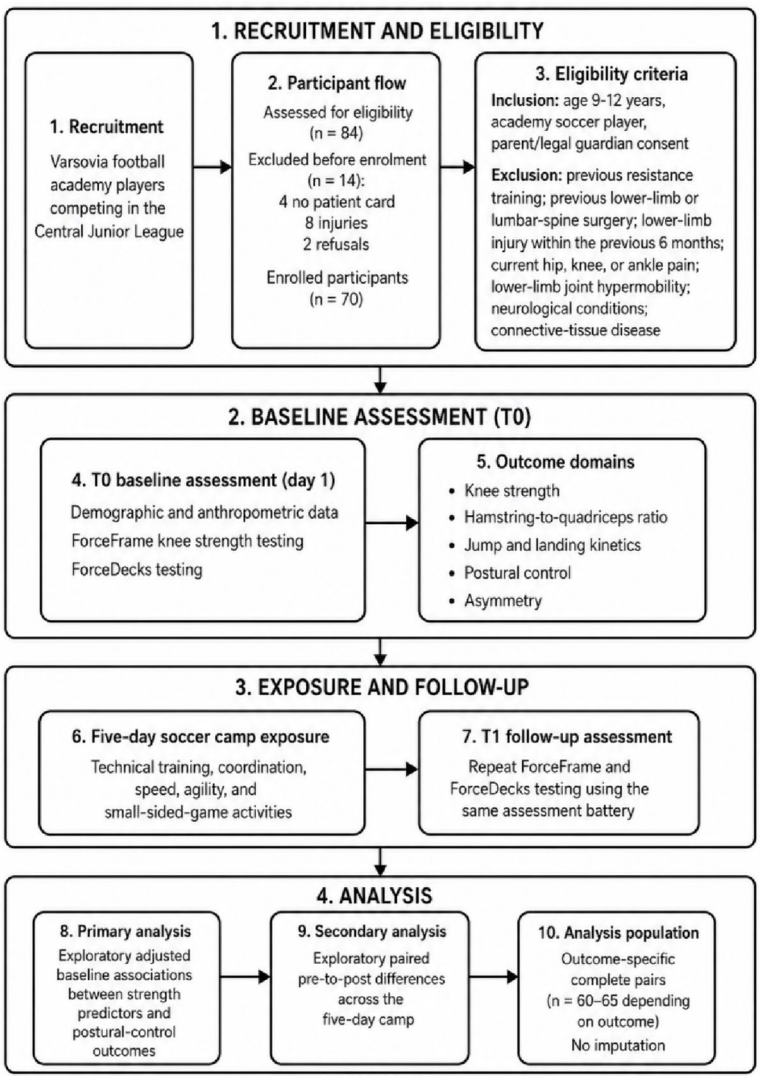


Fig. 1 Study design, recruitment, follow-up and analysis flow.

Participant characteristics, anthropometry, limb dominance, and maturity-related variables

Sex, recorded age group, and baseline body mass were available for the analysis. Age was stored as an integer age-group value from 9 to 12 years rather than exact chronological age calculated from date of birth and assessment date; it is therefore referred to as age group throughout the Methods and statistical models. Body mass was measured at T0 using the ForceDecks platforms while participants were barefoot and wore lightweight athletic clothing.

Maturity offset was estimated anthropometrically using the sex-specific equations developed by Mirwald et al.[19]. Stature and sitting height were measured in centimetres using calibrated anthropometric equipment, and body mass was recorded in kilograms with participants wearing light clothing and no footwear. Subischial leg length was calculated as standing stature minus sitting height, while chronological age was expressed in decimal years on the assessment date. For boys, maturity offset was calculated as: −9.236 + [0.0002708 × (leg length × sitting height)] − [0.001663 × (chronological age × leg length)] + [0.007216 × (chronological age × sitting height)] + [0.02292 × body-mass-to-stature ratio]; for girls, the corresponding equation was: −9.376 + [0.0001882 × (leg length × sitting height)] + [0.0022 × (chronological age × leg length)] + [0.005841 × (chronological age × sitting height)] − [0.002658 × (chronological age × body mass)] + [0.07693 × body-mass-to-stature ratio]. The body-mass-to-stature ratio was calculated as 100 × body mass (kg)/stature (cm). Maturity offset represents the estimated number of years from peak height velocity, with negative values indicating that a participant was estimated to be pre-peak height velocity and positive values indicating a post-peak height-velocity status; predicted age at peak height velocity was calculated by subtracting maturity offset from chronological age.

Limb dominance was defined as the preferred lower limb used for kicking a ball and was determined using question: which lower limb is the preferred lower limb for kicking a ball? The same definition was used for all side-order and direction-consistency analyses.

Male and female participants were retained in the cohort-level analyses because the study concerned the academy sample; however, the female subgroup was too small for stable sex-stratified estimation. Sex was included as a covariate in the primary baseline models, and a male-only sensitivity analysis examined the influence of the unequal sex distribution.

Sample-size rationale

The target sample was evaluated prospectively before recruitment using G*Power 3 (Faul et al. 2007). For a two-sided test of one correlation, alpha = 0.05, 80% power, and an assumed association of rho = 0.33, approximately 70 participants were required. This calculation used a single unadjusted correlation as an approximation for a Spearman association and did not account for covariate adjustment, the 18-member baseline analysis family, or false-discovery-rate control. The sample was therefore expected to provide limited precision for small associations and for multiplicity-adjusted estimates.

For the exploratory repeated-session component, complete-pair samples of approximately 60–65 participants provide about 80% power at two-sided alpha = 0.05 to detect a standardised paired difference of approximately d_z = 0.35–0.36. This calculation did not account for global correction across 24 outcomes and was not used to support confirmatory claims. No separate sample-size calculation supported sex-stratified analyses, interaction analyses, equivalence testing, or individual responder classification.

Assessment timing, pre-assessment control, and testing sequence

T0 was conducted between 08:30 and 11:00 on the first day of camp, before the observed camp exposure, whereas T1 was conducted between 15:30 and 18:00 on day 5. Participants completed no structured training overnight before T0. On day 5, the final training session ended at 11:00, the last meal before testing was provided at 13:00 with standardised content, and T1 commenced at 15:30. Participants were assessed in the same order at both time

points. Testing time and pre-assessment feeding conditions were not matched between T0 and T1; potential circadian and postprandial influences were therefore treated as uncontrolled measurement-session factors.

Pre-assessment instructions addressed strenuous exercise, caffeine, meals, hydration, medication, sleep, recent illness, pain, and recovery. No unverified recommendation was treated as a measured exposure.

Assessments were conducted in the same laboratory using an identical station order: ForceDecks assessments followed by ForceFrame pad-force testing. The fixed order was chosen to improve procedural consistency, although fatigue or potentiation caused by the preceding tasks could not be excluded. The complete battery required approximately 120 min per participant. Five minutes of recovery were provided between test stations.

Each assessment was preceded by a standardised 13-min warm-up comprising 5 min of jogging, 5 min of dynamic lower-limb mobility, and 3 min of potentiation exercises. The potentiation component included 10 bodyweight squats, 10 alternating lunges, 10 calf raises, three submaximal vertical jumps, and two maximal vertical jumps, with 20–30 s before the maximal attempts. The interval from completion of the warm-up to the first recorded test was 5 min and was reproduced at T1.

Participants wore their own footwear; footwear consistency between T0 and T1 was not documented. Lightweight clothing was standardised. Foot placement, stance width, and device settings were recorded at T0 and reproduced at T1 using floor markings and recorded foot spacing. The same examiner conducted the corresponding assessment at both time points.

The observed exposure was participation in a 5-day technical and tactical camp organised by the academy. It was not a research-assigned resistance, plyometric, balance, or neuromuscular intervention. The planned programme included ball control, dribbling, passing and receiving, shooting, coordination, speed and agility tasks, one-versus-one and three-versus-three small-sided games, task-based games, and practice matches. Eight 90-min sessions were scheduled per participant, corresponding to 720 planned minutes. Attendance was recorded by three lead coaches, two assistant coaches, one motor preparation coach, one physiotherapist, and completion was defined as participation for the full 90 min. Of the 70 participants, 65 completed all eight sessions. Five participants did not complete all sessions due to injuries. One participant discontinued the morning session on the fourth day after 60 min due to injury. In the afternoon session on the fourth day, four participants did not complete the protocol: three discontinued after 60 min, and one after 75 min. In all four cases, the reason for discontinuation was injuries.

The academy prescribed the same planned content across age groups, but neither internal load nor external load was measured. Accordingly, the study can describe planned activities and attendance but cannot establish comparable physiological or mechanical exposure between participants. Practice-match substitutions were managed by the coaching staff according to the goals of the given training unit. Individual playing time was not recorded, therefore it is not possible to determine or compare match exposure between players. A great representation of games were conducted using sided games as previous literature supports[20–23].

The one-versus-one format used a 15 × 10 m area, three 2-min bouts, and 2 min of recovery between bouts, providing 6 min of active play and 10 min including two recovery intervals. The three-versus-three format used a 30 × 20 m area, three 4-min bouts, and 2 min of recovery between bouts, providing 12 min of active play and 16 min including two recovery intervals. All games used size 4 balls and small goals (2 × 1 m). Several balls were placed around the field to allow for immediate restarts if the ball went out of bounds. Coaches provided standardized verbal encouragement during all games. Breaks between sets were for passive rest with opportunities for hydration. Identical organizational conditions and rules were used for all age groups (U9–U12). The practice match comprised two 25-min periods. Player substitutions were unlimited (rolling substitutions) and were made by the coaching staff according to the flow of the game and training objectives. Individual playing time was not recorded, and no comparisons were made between players.

Sleep, nutrition, hydration, additional physical activity, illness, pain, and recovery were not monitored using standardised objective tools. The academy provided three meals per day and unrestricted access to water. Sleep hours and meals were based on the sports camp schedule, but actual sleep time, hydration levels, and caffeine consumption were not monitored.

Five participants were reported as having injuries during the camp period. Two volunteers experienced pain in the lateral malleolus region of the ankle, two experienced pain in the tibial tuberosity region, and one experienced groin pain. In all cases, the pain was not a continuation of a previous injury. These events were assessed by the training staff, and the players' subsequent participation in activities was individually adjusted according to their current functional status and the training staff's recommendations. All injuries occurred on the fourth day of the football camp. All participants for whom complete data from the T1 assessment were available were included in the final analysis.

Injuries and other adverse events were monitored by the physiotherapist and the head coach. Following the injury, the players were monitored during subsequent training sessions to assess the persistence of symptoms and their ability to participate in football activities. The decision to return to full training was made individually based on the player's current functional status, reported symptoms, and ability to perform training tasks without significant limitations. Return to full training occurred after achieving adequate exercise tolerance and the ability to perform football-specific activities. This assessment was conducted by the coaching staff.

Testing personnel and masking

Testing was conducted by two assessors. Assessor 1 was a physiotherapist and research-centre employee with 3 years of experience using VALD systems and conducted the ForceFrame assessment. Assessor 2 was a physiotherapist and football coach with 3 years of VALD-system experience and conducted the ForceDecks assessment.

Participants and coaches were aware of the camp and testing schedule. Outcome assessors could not be blinded to visibly apparent characteristics such as approximate age or sex. During the T1 assessment, the individuals performing the measurements were blinded to the results obtained at the T0 assessment. Data from the T0 and T1 assessments were exported from the VALD system only after the data collection process was completed, so earlier results were not available when the T1 measurements were performed. T0 results were not displayed during T1 testing. The data analyst worked with an anonymized dataset in which participants were identified only by codes. Time point labels (T0 and T1) were not masked during data quality verification, data export from the VALD system, and statistical analysis.

Equipment, environment, calibration, and general standardisation

Isometric pad force was assessed using a ForceFrame Strength Testing System (VALD Performance, Brisbane, Australia; serial number FF-0672; ForceFrame iOS software version 1.6.1) at an acquisition frequency of 50 Hz. Force-platform assessments used two ForceDecks Lite plates (30 × 48.5 cm; serial numbers FDL-2-00812 and FDL-2-00631; ForceDecks iOS software version 1.0.0; hardware firmware FW2R4). Measurements on the ForceFrame platforms were recorded at a sampling rate of 1000 Hz.

Before each session, devices were placed on a rigid level surface and zeroed according to manufacturer procedures. ForceDecks were zeroed without contact before each participant, and ForceFrame pad position and frame geometry were checked against recorded marks.

Laboratory temperature was approximately 21 °C and relative humidity approximately 55%. Temperature and humidity were measured using a digital thermohygrometer before each testing session. The same laboratory layout, device position, instructions, rest intervals, side order, and trial-validity rules were used at T0 and T1.

Outcome framework and variable selection

The primary baseline analysis examined three ForceFrame-derived predictors: absolute bilateral mean maximum knee-flexion pad force, absolute bilateral mean maximum knee-extension pad force, and the protocol-specific knee-flexion-to-knee-extension pad-force ratio. The six ForceDecks-derived outcomes were selected to represent distinct constructs rather than a single homogeneous postural-control domain: single-leg-standing

centre-of-pressure medial-lateral range and anterior-posterior range represented sway amplitude; single-leg-standing centre-of-pressure mean velocity represented sway rate; mean-velocity asymmetry represented static interlimb asymmetry; single-leg land-and-hold time to stabilisation represented dynamic stabilisation; and single-leg land-and-hold peak-force asymmetry represented landing-load asymmetry. This set limited redundancy while retaining one or more measures from each prespecified construct of interest. No single association was treated as a confirmatory primary endpoint, and the complete 18-member family was included in multiplicity correction.

The secondary repeated-session component comprised 24 outcomes spanning ForceFrame pad force and asymmetry; countermovement-jump and squat-jump outputs; bilateral and single-leg landing force and time to stabilisation; single-leg-standing centre-of-pressure variables; and interlimb asymmetry. The full 24-outcome family was retained in the global false-discovery-rate correction.

Force was expressed in newtons (N), power relative to body mass in watts per kilogram (W/kg), displacement and jump height in centimetres (cm), time to stabilisation in seconds(s), centre-of-pressure range in millimetres (mm), centre-of-pressure mean velocity in millimetres per second (mm/s), and asymmetry in percentage points. Higher centre-of-pressure range or velocity indicates greater sway under the tested condition. Lower time to stabilisation indicates earlier fulfilment of the software-defined criterion; these directions are descriptive and do not establish clinical benefit.

ForceFrame isometric knee pad-force assessment

Maximum voluntary isometric knee-extension and knee-flexion pad force was assessed using ForceFrame. The system measured force applied to the dynamometer pad rather than knee-joint torque. For knee extension, participants stood on the contralateral limb while the tested knee was positioned at approximately 45° of flexion. The trunk was held upright, the pelvis was maintained in a neutral position, the hands gripped the ForceFrame handles, and the tested distal leg pressed against a pad positioned on the anterior distal tibia. For knee flexion, participants stood on the contralateral limb with the tested knee at approximately 30° of flexion and pressed the posterior distal leg against the pad. Knee angle was measured before each trial using a universal goniometer.

The pad was centred 2 cm proximal to the lateral malleolus. The perpendicular distance between the knee-joint axis and the pad was not measured; torque was therefore not calculated. The unilateral standing configuration may depend on balance, trunk stabilisation, contralateral-limb support, and coordination in addition to force production. Because lower-leg length and pad-to-axis distance were unavailable, body-mass adjustment could not remove all anthropometric influence on between-participant pad-force values.

Knee extension was tested before knee flexion. Each participant completed two submaximal practice contractions for each test mode, followed by three recorded maximal trials per limb. Each contraction lasted 3 s, with 30 s between trials and 1 min between limbs or test modes. Participants were instructed to increase force rapidly and maintain maximal effort. Standardised verbal encouragement and visual feedback from the measurement system were provided during both T0 and T1.

The knee-flexion-to-knee-extension pad-force ratio was calculated as 100 x bilateral mean knee-flexion pad force divided by bilateral mean knee-extension pad force. Flexion and extension were assessed in different standing positions and at different knee angles and force directions; the ratio was therefore treated as protocol specific and was not equated with an isokinetic or isometric torque ratio or interpreted as an injury-risk threshold.

ForceDecks assessments

ForceDecks were used to obtain vertical ground-reaction-force and centre-of-pressure outputs. Trial detection and phase identification were performed in the software version reported above. The analysis settings below must be reported explicitly because default algorithms can change between software versions. Adult validation evidence was used only to contextualise the device and was not assumed to establish paediatric reliability for every outcome[10].

Single-leg standing

Participants stood on one force plate on the tested limb, with arms alongside the body, the non-support limb held in slight knee flexion, and gaze directed at a fixed target located 2 m at eye level from the participant. Foot orientation and position were recorded and reproduced. Limbs were tested in order based on the operational dominance definition. One familiarisation trial and three recorded trials were completed per limb, with 30 s between trials.

Each recorded trial lasted 18 s after a 3-s pre-recording stabilisation period. The acquisition and processing frequency was 1000 Hz, and no additional downsampling was used. The centre-of-pressure signal was processed using. No additional investigator-applied filtering was used. Outcomes were medial-lateral range, anterior-posterior range, and mean velocity.

The retained side-specific value was selected using the average value of three valid trials generated automatically by ForceDecks software (VALD). All analyzed metrics were calculated from the same set of three trials.

Countermovement jump

Participants began upright with one foot on each plate, hands on hips, chest upright, and gaze forward. Stance width and foot angle were recorded and reproduced. After 3 s of quiet standing for body-weight determination, participants performed a maximal bilateral countermovement jump with self-selected depth while keeping their hands on their hips and landing under control. Three valid trials were completed with 30 s between trials; invalid trials were repeated after 60 s.

Jump height was calculated using the impulse-momentum method. Additional outcomes were peak power relative to body mass, reactive strength index modified, and peak landing force. A trial was considered valid if it was performed according to the study protocol, without loss of contact with the platform before the take-off phase, additional hop, or incorrect landing. All variables analyzed were from the same trial.

Squat jump

Participants performed maximal unloaded squat jumps with hands on hips, an upright trunk, and a starting knee angle of approximately 90°. Stance width was self-selected at T0, measured to the nearest 0.5 cm, and reproduced at T1. Participants were instructed to avoid a preparatory countermovement. Compliance was assessed by visual observation by the same experienced examiner. Objective force or time thresholds were not used, nor was automated software verification performed. Trials that failed to meet the performance criteria were repeated. Three valid trials were completed with 30 s between attempts.

Outcomes were peak force, maximum negative displacement, and concentric peak power relative to body mass. A trial was considered valid if it was performed according to the test instructions, without any apparent countermovement, loss of balance, or abnormal landing. The mean value of the three valid trials, automatically generated by ForceDecks software (VALD), was used for analysis. All analyzed results were from the same set of three trials.

Bilateral jump-and-land-and-hold test

Participants began on the force plates with one foot on each plate, feet approximately shoulder width apart, hands on hips, and gaze forward. They performed a maximal vertical jump, landed on both plates, and maintained the landing position for at least 3 s. This was a jump-and-land-and-hold task rather than a drop landing. Three valid trials were completed with 60 s between trials. Foot position and instructions were reproduced at T1.

The onset of landing was identified by a force threshold of 15 N, which had to be exceeded and maintained for at least 0.5 s. Stabilization time was calculated from the moment of landing initiation until a stable landing position was achieved according to the established analysis criteria. Signal analysis was performed using ForceDecks

software (VALD Performance), and the analysis settings used during data processing were consistent for all participants and both time points (T0 and T1). Trials in which a participant did not achieve the required stable landing position were considered invalid and repeated.

Single-leg land-and-hold test

Participants started with feet together 30 cm from the near edge of the plates, jumped from both legs, landed on the designated limb, and maintained the position for at least 3 s with hands on hips and the non-landing limb off the ground. The approach distance was measured to the nearest 0.5 cm and reproduced. Each limb completed three valid trials with 60 s between attempts in starting with the dominant limb.

A trial was valid when the participant landed with full foot contact on the designated limb, took no additional step, did not touch the ground with the opposite limb, maintained hands on hips, and held the position for the required period. Outcomes were time to stabilisation and peak landing force. Detection, filtering, analysis-window, and stabilisation settings followed the bilateral task. The result automatically generated by ForceDecks software (VALD Performance) was used for analysis. Both stabilization time and maximum landing force were derived from the same result generated by the software for a given limb. Side-specific values were retained for interlimb asymmetry and direction-consistency analyses.

Derived variables and interlimb asymmetry

Absolute interlimb asymmetry was calculated as 100 x absolute value of (L - R) divided by max(L, R), where L and R were the retained left- and right-limb values. The limb producing the larger value was recorded separately. Direction consistency was defined as the proportion retaining the same higher-value limb at T0 and T1 among participants without ties. Because absolute magnitude removes direction and may change when the higher-value limb switches, asymmetry magnitude and direction were interpreted separately (Parkinson et al. 2021).

Absolute pad force was used in the primary baseline models, with body mass included once as a covariate. Full-sample allometric residuals and force divided by body mass were evaluated as sensitivity analyses. Longitudinal force differences were reported in newtons because T1 body mass was unavailable; therefore, longitudinal N/kg or allometrically scaled changes were not calculated.

Data management, data quality, and missing observations

Each participant was assigned a unique study code. Raw and software-derived data were exported from the VALD systems and merged by participant code and time point. Data were exported by one investigator and independently verified by a second member of the research team. Discrepancies were resolved by rechecking the source data in VALD systems and reconciling the final decision between the investigators. Data-cleaning decisions were completed using predefined validity rules described above.

No outcome value was imputed. Baseline association samples were outcome specific, and repeated-session analyses required a valid T0–T1 pair for the relevant outcome. Missing or invalid observations were classified as: device error: 0; invalid study: 0; participant withdrawal: 0; pain: 0; injury: 5; organizational/scheduling problems: 0; other reasons: 0. All missing data were due to injuries that prevented participation in the T1 assessment.

Statistical procedures

Continuous participant characteristics were summarised as mean plus or minus standard deviation (SD) and range, and categorical characteristics as number and percentage. Baseline availability and longitudinal complete pairs were reported separately by outcome. Complete and incomplete analysis populations were described using group-specific means, SDs, counts, and Hedges g for age group, body mass, and the maturity-offset score;

inferential p values were not emphasised because several incomplete groups were small. The primary baseline analysis used outcome-specific available cases, with a common complete-case sensitivity analysis across all 18 associations. No outcome imputation was performed.

The primary baseline analysis comprised 18 associations between three ForceFrame predictors and six ForceDecks outcomes. Predictor, outcome, recorded age group, sex, baseline body mass, and age-group-centred maturity offset were rank transformed. The predictor and outcome were separately residualised on the ranked covariates, and the Pearson correlation between the residuals was reported as an adjusted partial Spearman coefficient (rho). Age-group-centred maturity offset was calculated by subtracting the corresponding age-group mean from each value. This model retained absolute pad force and included body mass once, avoiding the previous redundant combination of allometric residualisation and additional body-mass adjustment.

For each coefficient, the participant number, exact two-sided p value, and 95% confidence interval (CI) were reported. Participant-level bootstrap CIs used 10,000 resamples, with rank transformation and covariate residualisation refitted within each resample. Benjamini-Hochberg false-discovery-rate correction was applied across all 18 coefficients. An adjusted q value of less than 0.05 was interpreted as statistical robustness to this multiplicity family; q values of 0.05 or greater were interpreted as inconclusive rather than evidence of no association or equivalence.

Sensitivity analyses examined whether the baseline findings depended on the specification of developmental covariates. First, all 18 associations were repeated after omitting maturity offset while retaining recorded age group, sex, and baseline body mass. Additional models examined maturity offset without age group and the simultaneous inclusion of raw age group and raw maturity offset, together with alternative force-scaling, sex-composition, and sample-composition specifications.

For each of the 24 repeated-session outcomes, the paired difference was T1 minus T0. T0 and T1 means and SDs were calculated from the identical paired sample for each row. Difference-score distributions were evaluated using quantile-quantile plots, the Shapiro-Wilk test, and inspection for influential values. The primary estimate was the mean paired difference with a two-sided 95% CI and paired-samples t test. Cohen d_z was the mean paired difference divided by the SD of paired differences; bias-corrected and accelerated bootstrap 95% CIs used 10,000 paired participant-level resamples. The Wilcoxon signed-rank test was used as a non-parametric sensitivity analysis.

Benjamini-Hochberg correction was applied globally across all 24 repeated-session outcomes. Absolute mean differences were the principal estimates. For strictly positive scale variables, a relative difference was calculated as 100 x mean paired difference divided by the paired-sample T0 mean and was explicitly labelled as a difference relative to the paired-sample T0 mean rather than the mean of individual percentage changes. Ratio and asymmetry differences were reported in percentage points. No relative percentage was calculated for negative displacement.

An exploratory analysis examined whether each T1–T0 difference varied with age-group-centred maturity offset. For each outcome, a partial Spearman coefficient related the score to the paired difference after adjustment for recorded age group, sex, and baseline body mass. Participant-level bootstrap 95% CIs used 2000 resamples, and false-discovery-rate correction was applied across the 24 associations. These coefficients described heterogeneity in repeated-session differences and were not interpreted as causal effects of biological maturation.

Secondary robustness analyses compared maximum-based ForceFrame summaries with the available average-force summaries, compared paired t-test and Wilcoxon results, and described interlimb direction consistency. Adult measurement-error thresholds were not used to classify paediatric participants as responders, and the previous post hoc equivalence analysis based on plus or minus 0.20 baseline SD was omitted.

Analyses were performed in Python 3.13.5 using NumPy 2.3.5, SciPy 1.17.0, statsmodels 0.14.6, and Matplotlib 3.10.8. The bootstrap random seed was 20260729. Two-sided p and q values were reported to three decimal places, with values below 0.001 reported as less than 0.001. Estimates and 95% CIs were prioritised over threshold-only interpretation.

Results

The recorded age-group variable averaged 10.53 ± 1.24 years (range, 9–12), with 22 players in the 9-year group, 11 in the 10-year group, 15 in the 11-year group, and 22 in the 12-year group. Baseline body mass was 35.89 ± 5.75 kg (range, 28–48 kg). The maturity-offset estimate averaged − 3.78 ± 1.11 years from peak height velocity (range, − 5.78 to − 1.22 years). Raw maturity offset and age group were strongly correlated (Spearman ρ = 0.902); after centring within age group, the maturity-offset covariate had a mean of 0.00 ± 0.50 years (range, − 1.25 to 1.38 years). Figure 1 summarises participant and analysis flow. Eighty-four players were assessed for eligibility, 14 were excluded before enrolment (4 because of missing patient cards, 8 because of injuries, and 2 who declined participation), and 70 participants were enrolled.

Baseline availability ranged from 61 participants for countermovement-jump outcomes to 65 for knee-flexion pad force. Longitudinal complete-pair samples were 65 for knee flexion, 62 for knee extension and most squat-jump, bilateral land-and-hold, and single-leg-standing outcomes, 61 for countermovement-jump outcomes and squat-jump negative displacement, and 60 for single-leg land-and-hold outcomes (Table 1).

Primary baseline associations

Across the 18 baseline associations adjusted for recorded age group, sex, baseline body mass, and age-group-centred maturity offset, coefficients ranged from − 0.194 to 0.327 (Table 2; Fig. 2). No coefficient remained robust after false-discovery-rate correction (all q ≥ 0.162). The largest estimate was the association between knee-extension maximum pad force and single-leg-standing centre-of-pressure mean velocity (ρ = 0.327, 95% CI 0.059 to 0.557; $p = 0.012$; q = 0.162). Knee-flexion maximum pad force was also positively associated with single-leg land-and-hold time to stabilisation before correction (ρ = 0.310, 95% CI 0.066 to 0.512; $p = 0.018$; q = 0.162). The largest negative coefficient was observed for the flexion-to-extension pad-force ratio and single-leg land-and-hold peak-force asymmetry (ρ = − 0.194, 95% CI − 0.423 to 0.077; $p = 0.145$; q = 0.379). These estimates were therefore statistically inconclusive at the multiplicity-adjusted family level.

Secondary ForceFrame repeated-session differences

Knee-extension maximum pad force was higher at T1 by 28.86 N (95% CI 16.62 to 41.09), equivalent to 16.3% of the paired-sample T0 mean; d_z = 0.60 (95% CI 0.33 to 0.81), $p < 0.001$, and q < 0.001 (Table 3). Knee-flexion maximum pad force was higher by 9.05 N (95% CI 1.93 to 16.17; 7.1% of the paired-sample T0 mean; d_z = 0.31, 95% CI 0.04 to 0.57; $p = 0.013$), but this result was not robust after the 24-outcome correction (q = 0.108). The protocol-specific flexion-to-extension pad-force ratio was 5.67% points lower at T1 (95% CI − 10.47 to − 0.88; d_z = − 0.30, 95% CI − 0.53 to − 0.02; $p = 0.021$; q = 0.113). Flexion and extension interlimb asymmetry estimates were statistically inconclusive after correction.

Secondary jump, landing, and stabilisation outcomes

None of the countermovement-jump or squat-jump outcomes was robust after multiplicity correction (Table 4). Countermovement-jump height was 0.95 cm higher at T1 (95% CI − 0.46 to 2.36; $p = 0.182$; q = 0.381), while peak power relative to body mass and reactive strength index modified each produced small positive estimates with q = 0.267. Bilateral land-and-hold time to stabilisation was 0.054 s lower at T1 (95% CI − 0.100 to − 0.008; d_z = − 0.30, 95% CI − 0.55 to − 0.03; $p = 0.023$), but the adjusted q value was 0.113. Bilateral peak landing force was statistically inconclusive (mean difference 18.92 N, 95% CI − 103.76 to 141.60; q = 0.828).

Table 1 Participant characteristics and outcome-specific analysis populations.

Characteristic or outcome	Value/complete pairs, *n*
Recruited participants	70
Sex, male	62 (88.6%)
Sex, female	8 (11.4%)
Age (years)	10.53 ± 1.24; range 9–12
Body mass at T0 (kg)	35.89 ± 5.75; range 28–48
Maturity offset (years)	−3.78 ± 1.11; range − 5.78 to − 1.22
Knee flexion maximum force (N)	65
Knee extension maximum force (N)	62
Hamstring-to-quadriceps ratio (%)	62
Knee flexion interlimb force asymmetry (%)	65
Knee extension interlimb force asymmetry (%)	62
Countermovement jump height (cm)	61
Countermovement peak power per body mass (W/kg)	61
Countermovement reactive strength index modified (m/s)	61
Countermovement peak landing force (N)	61
Squat jump peak force (N)	62
Squat jump concentric peak power per body mass (W/kg)	62
Squat jump maximum negative displacement (cm)	61
Land-and-hold time to stabilisation (s)	62
Land-and-hold peak drop landing force (N)	62
Single-leg land-and-hold time to stabilisation (s)	60
Single-leg land-and-hold peak drop landing force (N)	60
Single-leg land-and-hold time-to-stabilisation asymmetry (%)	60
Single-leg land-and-hold peak-force asymmetry (%)	60
Single-leg standing centre-of-pressure medial-lateral range (mm)	62
Single-leg standing centre-of-pressure anterior-posterior range (mm)	62
Single-leg standing centre-of-pressure mean velocity (mm/s)	62
Single-leg standing medial-lateral range asymmetry (%)	62
Single-leg standing anterior-posterior range asymmetry (%)	62
Single-leg standing mean-velocity asymmetry (%)	62

Values are mean ± SD or n (%).

Single-leg stabilisation, centre-of-pressure, and asymmetry outcomes

Single-leg land-and-hold time to stabilisation was 0.062 s lower at T1 (95% CI − 0.121 to − 0.004; $p = 0.037$; q = 0.150), whereas peak landing force and both associated asymmetry magnitudes were statistically inconclusive (Table 5). During single-leg standing, centre-of-pressure mean velocity was 6.06 mm/s lower at T1 (95% CI − 10.81 to − 1.32; d_z = − 0.33, 95% CI − 0.57 to − 0.04; $p = 0.013$), but this estimate did not remain robust after global correction (q = 0.108). Medial–lateral and anterior–posterior centre-of-pressure ranges and all three standing asymmetry magnitudes were also statistically inconclusive after correction (Fig. 3).

Exploratory maturity-offset associations with repeated-session differences

After adjustment for recorded age group, sex, and baseline body mass, age-group-centred maturity offset was positively associated with the T1 − T0 difference in bilateral land-and-hold time to stabilisation (ρ = 0.484, 95% CI 0.268 to 0.660; $p < 0.001$; q = 0.002) and single-leg land-and-hold time to stabilisation (ρ = 0.380, 95% CI 0.124 to 0.585; $p = 0.004$; q = 0.043) (Table 6; Fig. 4). Positive coefficients indicate that players with a more advanced maturity-offset value within their age group tended to have a more positive T1 − T0 time-to-stabilisation difference. Associations with single-leg-standing medial-lateral range and mean velocity had unadjusted p values below 0.05 but did not remain robust after correction (both q = 0.071). None of the remaining maturity-offset associations with repeated-session differences had q < 0.05.

Table 2 Covariate-adjusted baseline associations between ForceFrame pad-force predictors and selected ForceDecks outcomes.

Predictor	Outcome	*n*	Adjusted ρ	95% CI	*p*	q
Flexion pad force	SLS CoP medial-lateral range	64	0.053	−0.231 to 0.332	0.685	0.738
Flexion pad force	SLS CoP anterior-posterior range	64	0.228	−0.074 to 0.491	0.080	0.360
Flexion pad force	SLS CoP mean velocity	64	0.173	−0.134 to 0.439	0.185	0.417
Flexion pad force	SLS CoP mean-velocity asymmetry	64	0.162	−0.120 to 0.424	0.217	0.434
Flexion pad force	SLLAH time to stabilisation	62	0.310	0.066 to 0.512	0.018	0.162
Flexion pad force	SLLAH peak-force asymmetry	62	−0.052	−0.336 to 0.254	0.697	0.738
Extension pad force	SLS CoP medial-lateral range	62	0.193	−0.080 to 0.454	0.147	0.379
Extension pad force	SLS CoP anterior-posterior range	62	0.148	−0.143 to 0.434	0.267	0.480
Extension pad force	SLS CoP mean velocity	62	0.327	0.059 to 0.557	0.012	0.162
Extension pad force	SLS CoP mean-velocity asymmetry	62	−0.024	−0.281 to 0.253	0.860	0.860
Extension pad force	SLLAH time to stabilisation	62	0.201	−0.043 to 0.427	0.130	0.379
Extension pad force	SLLAH peak-force asymmetry	62	0.266	−0.020 to 0.509	0.043	0.259
Flexion-to-extension pad-force ratio	SLS CoP medial-lateral range	62	−0.133	−0.410 to 0.159	0.318	0.498
Flexion-to-extension pad-force ratio	SLS CoP anterior-posterior range	62	0.130	−0.147 to 0.372	0.332	0.498
Flexion-to-extension pad-force ratio	SLS CoP mean velocity	62	−0.119	−0.401 to 0.178	0.375	0.519
Flexion-to-extension pad-force ratio	SLS CoP mean-velocity asymmetry	62	0.071	−0.209 to 0.319	0.595	0.714
Flexion-to-extension pad-force ratio	SLLAH time to stabilisation	62	0.099	−0.153 to 0.333	0.460	0.591
Flexion-to-extension pad-force ratio	SLLAH peak-force asymmetry	62	−0.194	−0.423 to 0.077	0.145	0.379

Predictors and outcomes were rank transformed and separately residualised on recorded age group, sex, baseline body mass, and age-group-centred maturity offset; the residual correlation is reported as adjusted ρ. The same absolute-force strategy was used for both pad-force predictors and no prior allometric residualisation was applied in the primary model. Participant-level bootstrap 95% CIs used 10,000 resamples with residualisation refitted in each resample. q values are Benjamini–Hochberg adjusted across all 18 associations. CI, confidence interval; CoP, centre of pressure; SLLAH, single-leg land-and-hold; SLS, single-leg standing.

Sensitivity and robustness analyses

The absence of a multiplicity-robust baseline association was consistent across all alternative developmental, scaling, and sample-composition models. Minimum adjusted q values were 0.871 for the age-only model, 0.456 for raw maturity offset without age group, 0.295 when raw age group and raw maturity offset were entered together, 0.207 for the full-sample allometric model, 0.268 for N/kg scaling, 0.435 for the male-only analysis, and 0.162 for the common complete-case analysis. Thus, the maturity-adjusted primary conclusion did not depend on one developmental covariate specification or force-scaling method.

The Wilcoxon sensitivity analysis also retained knee-extension maximum pad force as the only repeated-session outcome with $q < 0.05$. Reanalysis using bilateral mean average ForceFrame pad force gave a 9.28 N flexion difference (95% CI 2.91 to 15.65; $q = 0.007$ within the three-outcome aggregation-sensitivity family), a 27.47 N extension difference (95% CI 16.72 to 38.23; $q < 0.001$), and a −5.89%-point ratio difference (95% CI −10.70 to −1.08; $q = 0.017$). This sensitivity suggests that the direction of the ForceFrame findings was not solely a consequence of maximum-value aggregation; it does not alter the principal 24-outcome multiplicity result. The higher-value limb changed between T0 and T1 in 30.8% of knee-flexion and 30.6% of knee-extension pairs. Direction switching ranged from 33.3% to 48.4% across the available single-leg-standing and single-leg land-and-hold measures, supporting separate interpretation of asymmetry magnitude and direction.

Discussion

The primary analyses did not identify a multiplicity-robust association between isometric knee-flexion or knee-extension pad force, the protocol-specific flexion-to-extension pad force ratio, and the selected ForceDecks outcomes after adjustment for age group, sex, baseline body mass, and the age-group-centred maturity offset.

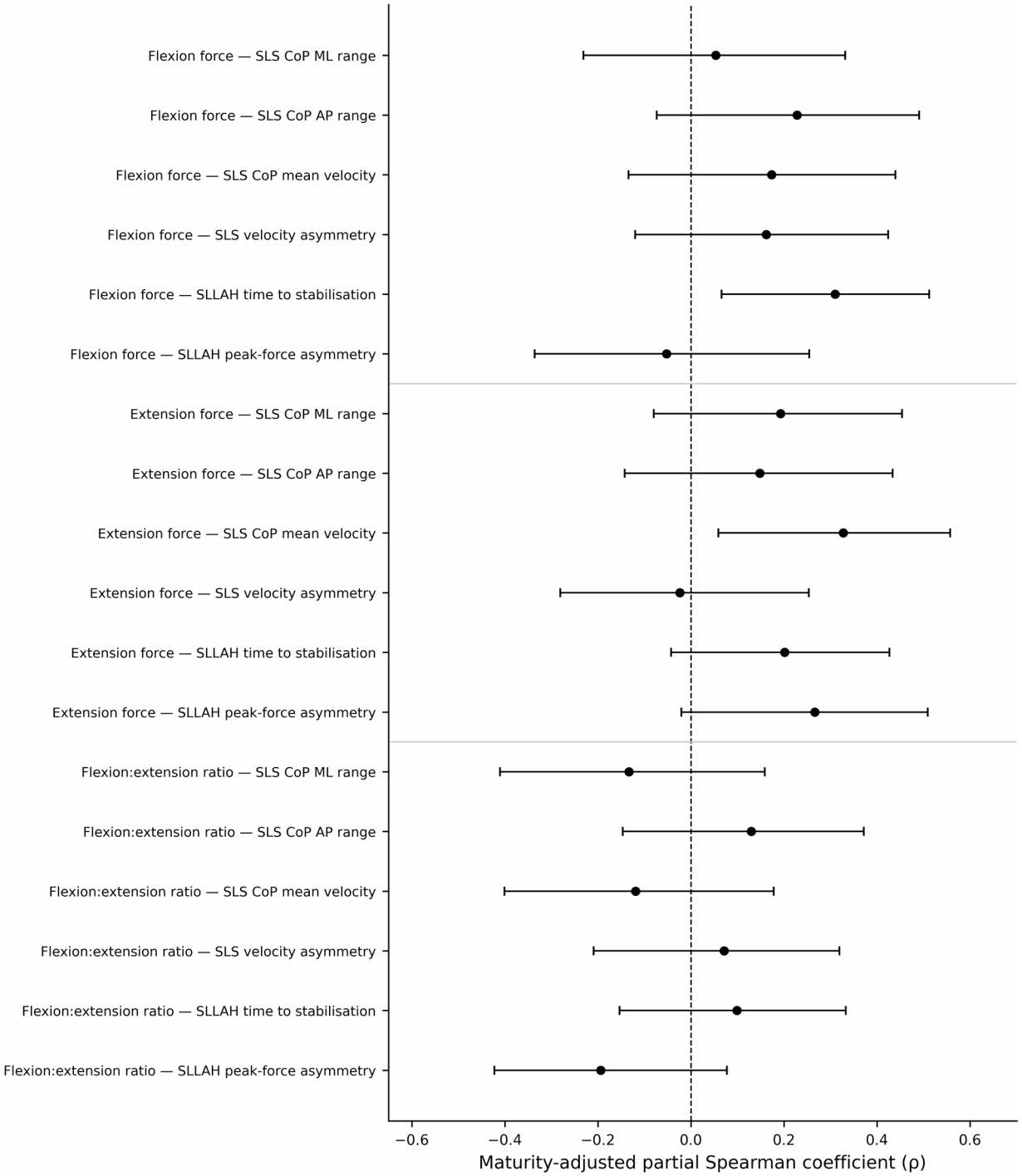


Fig. 2 Forest plot of the 18 covariate-adjusted primary baseline associations. Points are partial Spearman coefficients adjusted for age group, sex, body mass, and age-group-centred maturity offset and horizontal lines are participant-level bootstrap 95% CIs. The vertical line denotes $\rho = 0$. Sample size was 62–64 depending on outcome availability. No association had $q < 0.05$. *CoP* centre of pressure, *ML* medial–lateral, *AP* anterior–posterior, *SLLAH* single-leg land-and-hold, *SLS* single-leg standing.

Knee-extension maximum pad force was the only T1 − T0 difference that remained robust after global false-discovery-rate correction; all other repeated-session estimates were statistically inconclusive. These findings indicate that the assessed strength and postural-control measures were not interchangeable in this cohort, but they do not establish complementary clinical utility.

Table 3 ForceFrame repeated-session outcomes.

Outcome	Unit	*n*	T0, mean ± SD	T1, mean ± SD	T1 − T0 mean difference (95% CI)	*p*	q	d_z (95% CI)
Knee-flexion maximum pad force	N	65	127.43 ± 37.69	136.48 ± 39.54	9.05 (1.93 to 16.17)	0.013	0.108	0.32 (0.04 to 0.57)
Knee-extension maximum pad force	N	62	176.56 ± 54.21	205.42 ± 65.02	28.86 (16.62 to 41.09)	< 0.001	< 0.001	0.60 (0.33 to 0.81)
Knee-flexion-to-extension pad-force ratio	%	62	73.69 ± 19.37	68.02 ± 18.28	−5.67 (−10.46 to −0.88)	0.021	0.113	−0.30 (−0.53 to −0.02)
Knee-flexion interlimb pad-force asymmetry	%	65	12.73 ± 9.56	12.03 ± 9.10	−0.70 (−3.83 to 2.42)	0.654	0.826	−0.06 (−0.31 to 0.19)
Knee-extension interlimb pad-force asymmetry	%	62	16.08 ± 18.57	12.68 ± 9.15	−3.40 (−8.52 to 1.73)	0.190	0.381	−0.17 (−0.37 to 0.07)

All summaries use outcome-specific complete pairs. Ratio and asymmetry differences are percentage-point differences. The relative difference for pad force, when stated in text, is 100 × mean paired difference/paired-sample T0 mean. q values are adjusted across all 24 repeated-session outcomes, not only the five ForceFrame rows. *CI* confidence interva, *d_z* mean paired difference divided by the SD of paired differences, *T0* baseline, *T1* follow-up.

Table 4 Countermovement-jump, squat-jump, and bilateral land-and-hold repeated-session outcomes.

Outcome	Unit	*n*	T0, mean ± SD	T1, mean ± SD	T1 − T0 mean difference (95% CI)	*p*	q	d_z (95% CI)
Countermovement-jump height	cm	61	17.92 ± 4.41	18.87 ± 5.88	0.95 (−0.46 to 2.36)	0.182	0.381	0.17 (−0.09 to 0.37)
Countermovement-jump peak power relative to body mass	W/kg	61	35.86 ± 4.99	37.11 ± 6.42	1.24 (−0.24 to 2.73)	0.100	0.267	0.21 (−0.05 to 0.43)
Countermovement-jump reactive strength index modified	m/s	61	0.272 ± 0.081	0.286 ± 0.094	0.014 (−0.003 to 0.031)	0.100	0.267	0.21 (−0.05 to 0.46)
Countermovement-jump peak landing force	N	61	1745.34 ± 715.89	1675.59 ± 727.50	−69.75 (−220.61 to 81.10)	0.359	0.571	−0.12 (−0.36 to 0.15)
Squat-jump peak force	N	62	546.09 ± 138.01	557.04 ± 157.63	10.95 (−25.65 to 47.56)	0.552	0.779	0.08 (−0.19 to 0.32)
Squat-jump concentric peak power relative to body mass	W/kg	62	7.99 ± 2.50	8.40 ± 2.94	0.41 (−0.52 to 1.33)	0.381	0.571	0.11 (−0.16 to 0.34)
Squat-jump maximum negative displacement	cm	61	−34.81 ± 7.75	−36.16 ± 7.27	−1.35 (−3.27 to 0.57)	0.164	0.381	−0.18 (−0.42 to 0.09)
Bilateral land-and-hold time to stabilisation	s	62	0.528 ± 0.169	0.475 ± 0.143	−0.054 (−0.100 to −0.008)	0.023	0.113	−0.30 (−0.55 to −0.03)
Bilateral land-and-hold peak landing force	N	62	1528.47 ± 689.08	1547.39 ± 647.07	18.92 (−103.76 to 141.60)	0.759	0.828	0.04 (−0.21 to 0.29)

All summaries use outcome-specific complete pairs. Negative displacement is reported in centimetres without a relative percentage. q values are adjusted across all 24 repeated-session outcomes. Peak landing force refers to a maximal jump followed by landing. *CI* confidence interva, *d_z* standardised within-player difference, *T0* baseline, *T1* follow-up.

The small adjusted baseline coefficients are consistent with a systematic review and meta-analysis[7] reporting predominantly small relationships between balance and lower-extremity strength or power across the lifespan, with variation according to age and task. Soccer-specific evidence is not uniform since in an older male youth sample, isokinetic knee strength explained a modest proportion of variance in an overall stability index after accounting for age and body composition[13]. The difference from the present findings may reflect the younger and narrower age range, the use of maximal isometric rather than isokinetic strength, and the selection of conventional centre-of-pressure, stabilisation, and asymmetry outcomes rather than a single overall stability index. Biological maturity can influence static and dynamic postural control in male youth soccer players, and chronological age adjustment cannot substitute for direct assessment of maturity status[24]. Accordingly, the present null findings are best interpreted as an absence of strong adjusted relationships within the observed range, not as proof that strength and postural control are physiologically independent or clinically complementary. Restricted variability within one academy may also have attenuated associations and limits inference beyond similarly trained players.

Table 5 Single-leg land-and-hold and single-leg-standing repeated-session outcomes.

Outcome	Unit	*n*	T0, mean ± SD	T1, mean ± SD	T1 − T0 mean difference (95% CI)	*p*	q	d_z (95% CI)
Single-leg land-and-hold time to stabilisation	s	60	0.525 ± 0.185	0.463 ± 0.180	−0.062 (−0.121 to −0.004)	0.037	0.150	−0.27 (−0.53 to 0.00)
Single-leg land-and-hold peak landing force	N	60	970.25 ± 346.67	955.63 ± 363.70	−14.62 (−77.45 to 48.22)	0.643	0.826	−0.06 (−0.31 to 0.20)
Single-leg land-and-hold time-to-stabilisation asymmetry	%	60	31.66 ± 21.34	32.86 ± 19.48	1.20 (−6.58 to 8.98)	0.759	0.828	0.04 (−0.22 to 0.29)
Single-leg land-and-hold peak-force asymmetry	%	60	20.60 ± 14.02	18.37 ± 11.81	−2.23 (−6.79 to 2.33)	0.332	0.571	−0.13 (−0.38 to 0.14)
Single-leg-standing centre-of-pressure medial-lateral range	mm	62	32.34 ± 9.25	29.97 ± 7.94	−2.37 (−5.16 to 0.42)	0.095	0.267	−0.22 (−0.43 to 0.06)
Single-leg-standing centre-of-pressure anterior-posterior range	mm	62	53.71 ± 21.01	52.53 ± 20.61	−1.18 (−8.77 to 6.42)	0.758	0.828	−0.04 (−0.30 to 0.21)
Single-leg-standing centre-of-pressure mean velocity	mm/s	62	59.18 ± 18.24	53.11 ± 15.33	−6.06 (−10.81 to −1.32)	0.013	0.108	−0.32 (−0.57 to −0.04)
Single-leg-standing medial-lateral range asymmetry	%	62	22.88 ± 17.51	19.70 ± 14.98	−3.18 (−9.88 to 3.52)	0.346	0.571	−0.12 (−0.37 to 0.13)
Single-leg-standing anterior-posterior range asymmetry	%	62	28.58 ± 18.34	27.98 ± 17.36	−0.60 (−6.66 to 5.46)	0.843	0.880	−0.03 (−0.27 to 0.22)
Single-leg-standing mean-velocity asymmetry	%	62	18.71 ± 16.42	19.05 ± 14.11	0.33 (−5.61 to 6.28)	0.911	0.911	0.01 (−0.24 to 0.27)

All summaries use outcome-specific complete pairs. Ratio-like asymmetry outcomes are expressed as percentage-point differences. q values are adjusted across all 24 repeated-session outcomes. *CoP* centre of pressure, *CI* confidence interval, *d_z* standardised within-player difference, *T0* baseline, *T1* follow-up.

The only globally robust repeated-session finding was a 28.86 N (16.3%) higher knee-extension pad force at T1. ForceFrame measurement-error estimates from adult or older adolescent samples using non-identical protocols cannot be applied as paediatric thresholds in the present setting[9]. In the absence of a non-camp comparator, the observed difference cannot be separated from familiarisation, day-to-day biological variability, recovery status, measurement error, or other session-dependent influences. Familiarisation is a plausible contributor, but it was not directly evaluated in the present study[25]. Recent 4- to 6-week transitional programmes in older football players have produced mixed and domain-specific pre-to-post responses, including unchanged or reduced quadriceps and jump performance, which further illustrates that repeated-measures differences do not necessarily represent uniform physiological adaptation[26,27].

The hamstring-to-quadriceps ratio decrease was not robust after global multiplicity correction and was mathematically sensitive to the larger extension denominator. It should therefore be interpreted as an uncertain ratio difference across two testing sessions, not as evidence of selective quadriceps adaptation, hamstring insufficiency, or increased injury risk. The clinical meaning of a hamstring-to-quadriceps ratio depends on contraction mode, angular velocity, joint position, and protocol, and critical reviews do not support the ratio as an independent stand-alone predictor of hamstring or anterior cruciate ligament injury[28]. The present isometric ratio, obtained over 5 days without prospective injury surveillance, is therefore unsuitable for risk classification. The asymmetry findings reinforce the same caution: no absolute asymmetry magnitude changed robustly, and the limb producing the larger value switched in approximately one-third to one-half of repeated tests. Youth soccer studies similarly report task-specific asymmetry magnitude and unstable direction, and recent work found only small or inconsistent relationships between single-leg jump and isokinetic strength asymmetry[29,30]. Absolute percentage asymmetry consequently obscures directional information and should not be treated as a stable injury-related construct in isolation. Any recommendation to monitor relative hamstring development must remain tentative and should depend on a reproducible pattern obtained after adequate familiarisation, stable testing conditions, and paediatric reliability thresholds rather than on the present ratio difference alone.

https://doi.org/10.1038/s41598-026-69517-z

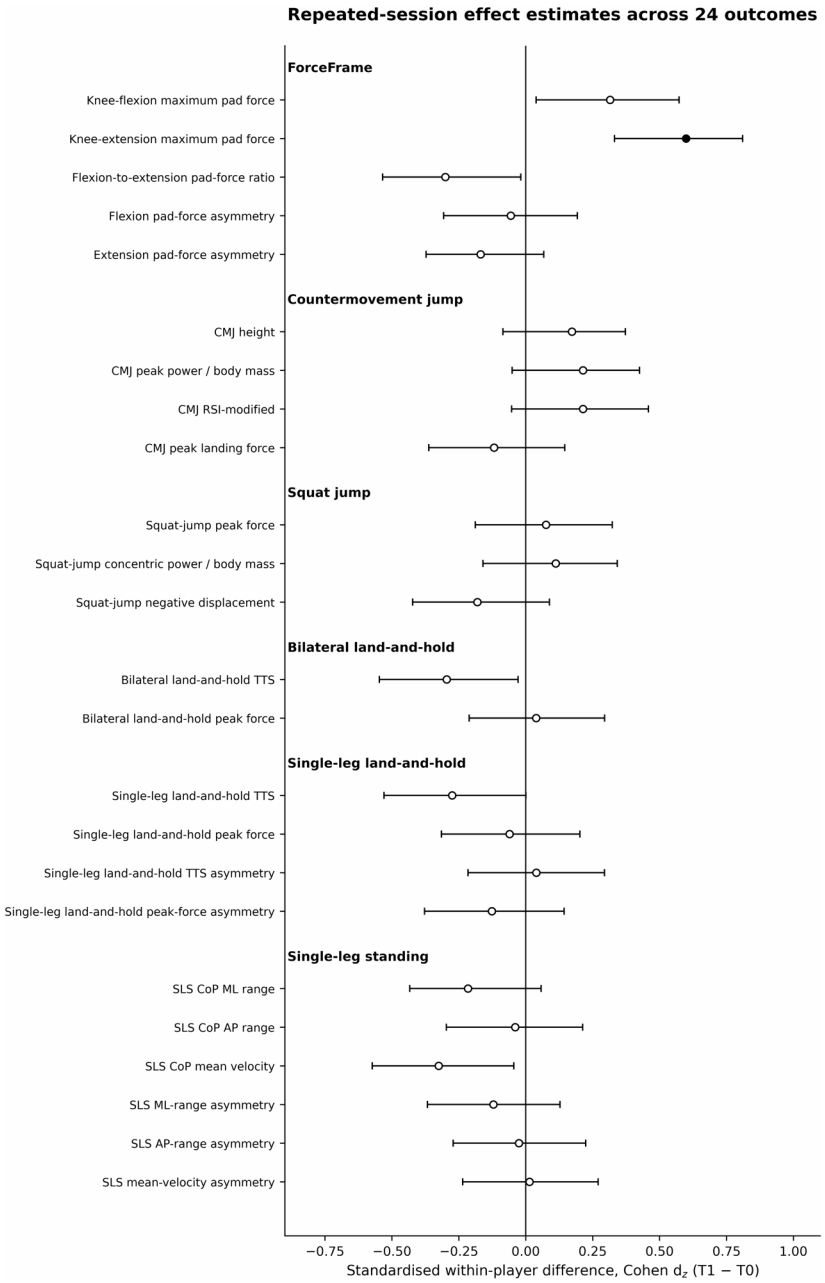


Fig. 3 Forest plot of standardised within-player repeated-session differences across 24 outcomes. Points are Cohen d_z estimates and horizontal lines are bootstrap 95% CIs. The filled point identifies the only outcome with $q < 0.05$ after correction across all 24 outcomes. A negative estimate indicates a lower value at T1; the direction is descriptive and does not imply clinical benefit. Sample size varied by outcome. *CoP* centre of pressure, *CMJ* countermovement jump, *ML* medial–lateral, *AP* anterior–posterior, *RSI* reactive strength index, *SLS* single-leg standing, *TTS* time to stabilisation.

The jump, landing, and conventional centre-of-pressure outcomes provided no additional multiplicity-adjusted findings. This does not demonstrate an absence of change. Several confidence intervals remained compatible with small potentially meaningful increases or decreases, and post hoc equivalence was supported only for the bilateral and single-leg landing-force outcomes. The interpretation is also measurement-specific because conventional centre-of-pressure variables do not capture every aspect of sport-related postural regulation. In young trained athletes, horizontal ground-reaction-force temporal metrics differentiated trained and untrained groups more clearly than selected spatial centre-of-pressure metrics, indicating that the two signal families can provide different information[31]. Comparisons among elite athletes, football players, and non-athletic controls have also shown that conventional centre-of-pressure differences are not uniformly favourable to football players, underscoring the task- and sport-specific nature of postural-control assessment[32]. The present results therefore apply to the selected ForceDecks outputs rather than to postural control as a unitary construct. Future analyses should prespecify a

Table 6 Associations between age-group-centred maturity offset and repeated-session differences.

Domain	Outcome	*n*	Adjusted ρ	95% CI	*p*	q
ForceFrame	Knee-flexion maximum pad force	65	−0.194	−0.398 to 0.042	0.130	0.446
ForceFrame	Knee-extension maximum pad force	62	0.040	−0.214 to 0.272	0.762	0.871
ForceFrame	Knee-flexion-to-extension pad-force ratio	62	−0.205	−0.439 to 0.064	0.119	0.446
ForceFrame	Knee-flexion interlimb pad-force asymmetry	65	0.098	−0.162 to 0.357	0.449	0.741
ForceFrame	Knee-extension interlimb pad-force asymmetry	62	0.075	−0.184 to 0.349	0.574	0.788
CMJ	Countermovement-jump height	61	0.042	−0.226 to 0.335	0.755	0.871
CMJ	Countermovement-jump peak power relative to body mass	61	0.013	−0.256 to 0.294	0.926	0.926
CMJ	Countermovement-jump reactive strength index modified	61	−0.155	−0.388 to 0.109	0.245	0.544
CMJ	Countermovement-jump peak landing force	61	−0.178	−0.416 to 0.086	0.181	0.544
SQT	Squat-jump peak force	62	−0.163	−0.384 to 0.074	0.218	0.544
SQT	Squat-jump concentric peak power relative to body mass	62	−0.251	−0.488 to −0.014	0.055	0.264
SQT	Squat-jump maximum negative displacement	61	0.154	−0.125 to 0.434	0.249	0.544
LAH	Bilateral land-and-hold time to stabilisation	62	0.484	0.268 to 0.660	<0.001	0.002
LAH	Bilateral land-and-hold peak landing force	62	−0.042	−0.298 to 0.260	0.752	0.871
SLLAH	Single-leg land-and-hold time to stabilisation	60	0.380	0.124 to 0.585	0.004	0.043
SLLAH	Single-leg land-and-hold peak landing force	60	−0.113	−0.398 to 0.192	0.404	0.741
SLLAH	Single-leg land-and-hold time-to-stabilisation asymmetry	60	0.099	−0.148 to 0.362	0.463	0.741
SLLAH	Single-leg land-and-hold peak-force asymmetry	60	0.032	−0.259 to 0.280	0.813	0.887
SLS	Single-leg-standing centre-of-pressure medial-lateral range	62	0.330	0.078 to 0.536	0.011	0.071
SLS	Single-leg-standing centre-of-pressure anterior-posterior range	62	0.021	−0.244 to 0.289	0.872	0.910
SLS	Single-leg-standing centre-of-pressure mean velocity	62	0.326	0.094 to 0.529	0.012	0.071
SLS	Single-leg-standing medial-lateral range asymmetry	62	0.071	−0.187 to 0.346	0.591	0.788
SLS	Single-leg-standing anterior-posterior range asymmetry	62	0.077	−0.202 to 0.379	0.562	0.788
SLS	Single-leg-standing mean-velocity asymmetry	62	0.126	−0.121 to 0.372	0.342	0.685

The coefficient relates age-group-centred maturity offset to the T1 − T0 paired difference after adjustment for recorded age group, sex, and baseline body mass. Positive coefficients denote more positive change scores with a more advanced maturity-offset value within age group. Bootstrap 95% CIs used 2000 participant-level resamples. q values are adjusted across all 24 associations. These analyses are exploratory and do not estimate a causal effect of maturation.

limited set of primary variables or validated composite scores and may consider dimension-reduction approaches, alongside horizontal ground-reaction-force, temporal, nonlinear, and more ecologically demanding balance measures, to reduce multiplicity while preserving construct coverage.

Limitations and future research

The longitudinal component lacked a non-camp comparison group, and repeated-session differences therefore remain descriptive. A further important limitation is that assessment time was systematically different between sessions: T0 was conducted in the morning (08:30–11:00), whereas T1 was conducted in the afternoon (15:30–18:00). Time-of-day effects have been demonstrated in young soccer players for physical-performance outcomes and have also been reported for postural-control measures. In addition, T1 followed a standardised meal provided at 13:00, whereas an equivalent pre-test feeding interval was not documented for T0. Circadian phase and feeding state were therefore structurally confounded with assessment session, and their contributions to the T1–T0 differences cannot be separated from familiarisation, recovery status, measurement variability, or the intervening camp exposure. This limitation is particularly relevant when interpreting the higher knee-extension pad force at T1 and further precludes attribution of the repeated-session differences to physiological adaptation. Training load, sleep, hydration, fatigue, and recovery were not quantified, and one familiarisation trial may have been insufficient for all tasks.

https://doi.org/10.1038/s41598-026-69517-z

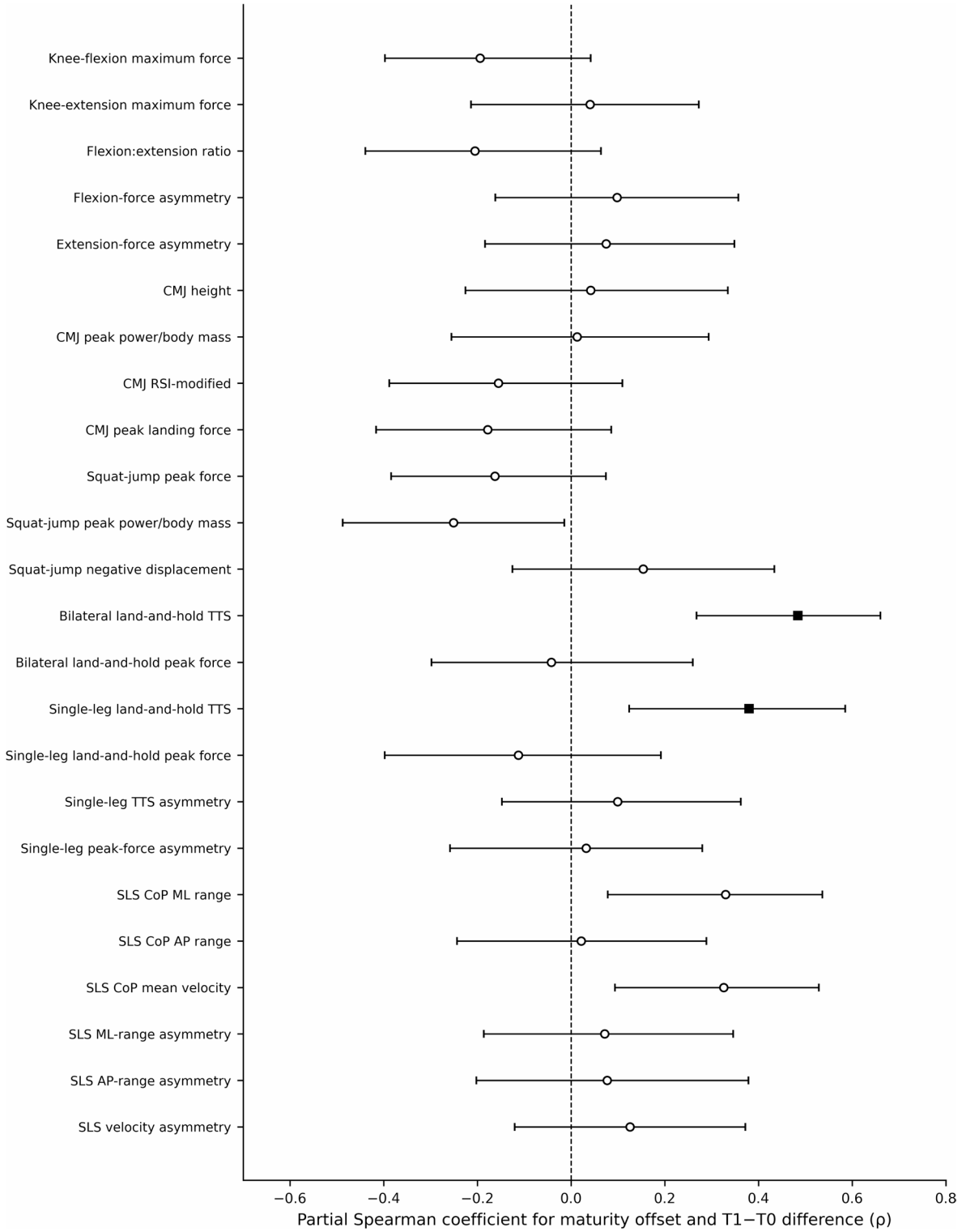


Fig. 4 Forest plot of maturity-offset associations with repeated-session differences.

Practical applications

For practitioners working with players of similar age and training status, maximal knee force, the hamstring-to-quadriceps ratio, conventional centre-of-pressure metrics, landing variables, and absolute asymmetry should not be substituted for one another or used individually to classify injury risk. The present data support measuring each characteristic only when it addresses a defined monitoring question, while recognising that the study did not establish the independent practical value of combining them. Serial monitoring should use identical positioning, instructions, assessor procedures, timing, and recovery conditions, with more than one familiarisation or baseline session when feasible and interpretation against paediatric, task-specific measurement-error thresholds. The isolated T1 increase in knee extension and the corresponding ratio decrease do not justify a change in training prescription. At most, a similar pattern reproduced across standardised sessions would warrant separate reassessment of flexor and extensor force rather than interpretation of the ratio alone. Injury-prevention decisions should remain based on prospective history, growth and maturation, workload and exposure, symptoms, and multidomain clinical and performance information rather than on a single ForceFrame or ForceDecks value.

Conclusion

In academy soccer players aged 9–12 years, absolute isometric knee-flexion and knee-extension pad force and the protocol-specific flexion-to-extension pad-force ratio showed no multiplicity-robust associations with the selected ForceDecks outcomes after adjustment for age group, sex, body mass, and age-group-centred maturity offset. Higher knee extension scores were observed at T1, whereas most other repeated-session differences were statistically and practically inconclusive and individual responses were heterogeneous; without a control group and paediatric measurement-error thresholds, these findings should be interpreted as uncertain measurement-session variation rather than camp-induced adaptation. The principal practical implication is that isolated strength, ratio, centre-of-pressure, landing, and asymmetry values should neither substitute for one another nor be used alone for injury-risk classification, and that longitudinal monitoring in this age group requires rigorous standardisation, adequate familiarisation, direct consideration of maturation, and paediatric reliability data.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-69517-z.

Author contributions R.S wrote the main manuscript text and prepared figures.

Funding No sources of funding were used to assist in the preparation of this article.

Data availability All data generated or analyzed during this study are included in this published article.

Declarations

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Mandorino, M., Figueiredo, J., Gjaka, A., Tessitore, A. & M., & Injury incidence and risk factors in youth soccer players: a systematic literature review. Part I: epidemiological analysis. *Biology Sport*. **40** (1), 3–25. https://doi.org/10.5114/biolsport.2023.109961 (2023a).
2. Mandorino, M., Figueiredo, J., Gjaka, A., Tessitore, A. & M., & Injury incidence and risk factors in youth soccer players: a systematic literature review. Part II: Intrinsic and extrinsic risk factors. *Biology Sport*. **40** (1), 27–49. https://doi.org/10.5114/biolsport.2023.109962 (2023b).
3. Clemente, F. M. et al. Exploring relationships between anthropometry, body composition, maturation, and selection for competition: a study in youth soccer players. *Front. Physiol.* https://doi.org/10.3389/fphys.2021.651735 (2021).
4. Ishøi, L. et al. Hamstring and quadriceps muscle strength in youth to senior elite soccer: a cross-sectional study including 125 players. *Int. J. Sports Physiol. Perform.* **16** (10), 1538–1544. https://doi.org/10.1123/ijspp.2020-0713 (2021).
5. Read, P. J., Oliver, J. L., De Ste Croix, M. B. A., Myer, G. D. & Lloyd, R. S. Neuromuscular risk factors for knee and ankle ligament injuries in male youth soccer players. *Sports Med.* **46** (8), 1059–1066. https://doi.org/10.1007/s40279-016-0479-z (2016).
6. Baroni, B. M., Ruas, C. V., Ribeiro-Alvares, J. B. & Pinto, R. S. Hamstring-to-quadriceps torque ratios of professional male soccer players: a systematic review. *J. Strength. Conditioning Res.* **34** (1), 281–293. https://doi.org/10.1519/JSC.0000000000002609 (2020).
7. Muehlbauer, T., Gollhofer, A. & Granacher, U. Associations between measures of balance and lower-extremity muscle strength/power in healthy individuals across the lifespan: a systematic review and meta-analysis. *Sports Med.* **45** (12), 1671–1692. https://doi.org/10.1007/s40279-015-0390-z (2015a).
8. Pau, M. et al. Relationship between static and dynamic balance abilities in Italian professional and youth league soccer players. *Phys. Ther. Sport*. **16** (3), 236–241. https://doi.org/10.1016/j.ptsp.2014.12.003 (2015).
9. Arp, K. et al. Validity, agreement and reliability of the ForceFrame dynamometer in patients with anterior cruciate ligament injury. *Int. J. Sports Phys. Therapy*. https://doi.org/10.26603/001c.122486 (2024).
10. Collings, T. J., Lima, Y. L., Dutaillis, B. & Bourne, M. N. Concurrent validity and test–retest reliability of VALD ForceDecks' strength, balance, and movement assessment tests. *J. Sci. Med. Sport*. **27** (8), 572–580. https://doi.org/10.1016/j.jsams.2024.04.014 (2024).
11. Dhahbi, W. & Chamari, K. The algorithmic athlete: a call to standardize assessment of sensor technologies and artificial intelligence. *Int. J. Sports Physiol. Perform.* **21** (4), 505–506. https://doi.org/10.1123/ijspp.2025-0547 (2026).
12. Towlson, C. et al. Maturity-associated considerations for training load, injury risk, and physical performance in youth soccer: one size does not fit all. *J. Sport Health Sci.* **10** (4), 403–412. https://doi.org/10.1016/j.jshs.2020.09.003 (2021).
13. França, C. et al. Knee muscle strength, body composition, and balance performance of youth soccer players. *BMC Sports Sci. Med. Rehabilitation*. **16** (1), 206. https://doi.org/10.1186/s13102-024-00966-7 (2024).
14. Lesinski, M., Prieske, O. & Granacher, U. Effects and dose–response relationships of resistance training on physical performance in youth athletes: a systematic review and meta-analysis. *Br. J. Sports Med.* **50** (13), 781–795. https://doi.org/10.1136/bjsports-2015-095497 (2016).
15. Sánchez Pastor, A., García-Sánchez, C., Nieto, M., de la Rubia, A. & M. Influence of strength training variables on neuromuscular and morphological adaptations in prepubertal children: a systematic review. *Int. J. Environ. Res. Public Health*. **20** (6), 4833. https://doi.org/10.3390/ijerph20064833 (2023).
16. Brito, J. et al. Postural stability decreases in elite young soccer players after a competitive soccer match. *Phys. Ther. Sport*. **13** (3), 175–179. https://doi.org/10.1016/j.ptsp.2011.07.004 (2012).
17. Pau, M., Ibba, G. & Attene, G. Fatigue-induced balance impairment in young soccer players. *J. Athl. Train.* **49** (4), 454–461. https://doi.org/10.4085/1062-6050-49.2.12 (2014).
18. Yamada, R. K. F. et al. The effects of one-half of a soccer match on the postural stability and functional capacity of the lower limbs in young soccer players. *Clinics* **67** (12), 1361–1364. https://doi.org/10.6061/clinics/2012(12)03 (2012).
19. Mirwald, R. L. R. L. et al. An assessment of maturity from anthropometric measurements. *Med. Sci. Sports Exerc.* **34** (4), 689–694. https://doi.org/10.1097/00005768-200204000-00020 (2002).
20. Clemente, F. M. et al. The effects of small-sided soccer games on tactical behavior and collective dynamics: a systematic review. *Chaos Solitons Fractals*. https://doi.org/10.1016/j.chaos.2020.109710 (2020).
21. Clemente, F. M., Afonso, J. & Sarmento, H. Small-sided games: an umbrella review of systematic reviews and meta-analyses. *PLoS One*. **16** (2), e0247067. https://doi.org/10.1371/journal.pone.0247067 (2021).
22. Clemente, F. M. et al. Variability of internal and external loads and technical/tactical outcomes during small-sided soccer games: a systematic review. *Biology Sport*. **3**, 647–672. https://doi.org/10.5114/biolsport.2022.107016 (2022).
23. Clemente, F. et al. Effects of pitch size on soccer players' physiological, physical, technical, and tactical responses during small-sided games: a meta-analytical comparison. *Biology Sport*. **40** (1), 111–147. https://doi.org/10.5114/biolsport.2023.110748 (2023).
24. John, C., Rahlf, A. L., Hamacher, D. & Zech, A. Influence of biological maturity on static and dynamic postural control among male youth soccer players. *Gait Posture*. **68**, 18–22. https://doi.org/10.1016/j.gaitpost.2018.10.036 (2019).

25. McGoldrick, C. D. et al. Reliability, familiarization effect, and comparisons between a predetermined and a self-determined isometric-squat testing protocol. *Int. J. Sports Physiol. Perform.* **18** (7), 718–725. https://doi.org/10.1123/ijspp.2022-0480 (2023).
26. Isbilir, M. et al. Pre- and post-test evaluation of a periodized off-season training program in professional footballers. *Appl. Sci.* **15** (19), 10354. https://doi.org/10.3390/app151910354 (2025).
27. Michailidis, Y. et al. The effect of a high-frequency exercise program during the transition period in young football players. *Sports* **13** (9), 297. https://doi.org/10.3390/sports13090297 (2025).
28. Kellis, E., Sahinis, C. & Baltzopoulos, V. Is hamstrings-to-quadriceps torque ratio useful for predicting anterior cruciate ligament and hamstring injuries? A systematic and critical review. *J. Sport Health Sci.* **12** (3), 343–358. https://doi.org/10.1016/j.jshs.2022.01.002 (2023).
29. Ascenzi, G. et al. Inter-limb asymmetry in youth elite soccer players: effect of loading conditions. *PLOS ONE*. **17** (6), e0269695. https://doi.org/10.1371/journal.pone.0269695 (2022).
30. Michailidis, Y. et al. Correlation of the asymmetry index from the single-leg countermovement jump with the asymmetry index from isokinetic strength in elite youth football players. *Appl. Sci.* **15** (5), 2779. https://doi.org/10.3390/app15052779 (2025).
31. Borzucka, D., Kręcisz, K. & Kuczyński, M. Ground reaction forces better than center of pressure differentiate postural control between young female volleyball players and untrained peers. *Sci. Rep.* **14** (1), 5869. https://doi.org/10.1038/s41598-024-56398-3 (2024).
32. Wojciechowska-Maszkowska, B., Borzucka, D. & Rogowska, A. M. Comparison of balance skills, personality, and temperament of elite sports athletes and football players. *J. Phys. Educ. Sport*. **20** (6), 3671–3683 (2020).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Rafał Studnicki[1]

✉ Rafał Studnicki
rafal.studnicki@gumed.edu.pl

[1] Department of Physiotherapy, Medical University of Gdańsk, 7 Dębinki Street, 80-211 Gdańsk, Poland